ModelIt!

Teacher's Guide

Nature's Recycling System

Why Earth Doesn't Fill Up with Dead Stuff

Quick Reference

Grade Level	Grade 5
NGSS Standard	5-LS2-1
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: 5-LS2-1

Develop a model to describe the movement of matter among plants, animals, decomposers, and the environment.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a model to describe matter cycling through decomposers.
Disciplinary Core Idea	**LS2.B Cycles of Matter and Energy Transfer** — Matter cycles between living and nonliving parts of an ecosystem.
Crosscutting Concept	**Systems and System Models** — Students model the ecosystem as a system with inputs and outputs.

CAST Testing Alignment

MULTIPLE CHOICE:

When soil moisture decreases, what happens to decomposer activity?

CONSTRUCTED RESPONSE:

Explain how dead leaves eventually become nutrients that plants can use.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Decomposition is the process by which dead organic matter is broken down into simpler substances. This process is essential for nutrient cycling and soil health.

The Decomposer Community

Decomposers include bacteria, fungi, and invertebrates like earthworms. Fungi are particularly important because they can break down tough materials like lignin in wood.

Moisture Requirements

Decomposers need moisture to survive and transport nutrients. In dry conditions, bacteria become dormant and fungi retreat, dramatically slowing decomposition.

Temperature Effects

Decomposition speeds up in warm temperatures and slows in cold. This explains why leaves accumulate on forest floors in winter but quickly disappear in summer.

The Nitrogen Cycle

Decomposers play a crucial role in the nitrogen cycle by breaking down proteins in dead matter and releasing nitrogen compounds that plants can absorb.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L — **Locate the System**
Students identify decomposers, dead organic matter, soil moisture, and nutrient release as system components.

E — **Establish Relationships**
Students determine that moisture and organic matter have positive effects on decomposer activity.

V — **Visualize & Model**
Students build the decomposition model showing how inputs drive activity and output.

E — **Evaluate Outcomes**
Students run drought scenarios and observe the cascade of effects on nutrient release.

R — **Revise & Extend**
Students add temperature or other factors to make the model more realistic.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

1 min

Say: *"Today we're exploring nature's recycling system."*

Watch For

Generate curiosity with the question: Why doesn't Earth fill up with dead stuff?

Slide 2: Learning Objectives

2 min

Say: *"These are our learning goals for today."*

Watch For

Have students read objectives aloud.

Slide 3: Big Question

2 min

Say: *"This is the question our model will help us answer."*

Watch For

Poll students for initial ideas.

Slide 4: LEVER Framework

2 min

Say: *"We'll use the LEVER framework to build our model."*

Watch For

Briefly explain each step.

Slide 5: Activity 1: Components

8 min

Say: *"Let's identify the parts of our decomposition system."*

Watch For

Guide students through sorting components.

Slide 6: Activity 2: Connections

8 min

Say: *"Now let's connect our components with positive and negative relationships."*

Watch For

Demonstrate how to draw arrows in ModelIt.

Slide 7: Activity 3: Simulation

10 min

Say: *"Let's test our model by running simulations."*

Watch For

Have students run drought scenario and record observations.

Slide 8: Discoveries

5 min

Say: *"What did our model reveal about decomposition?"*

Watch For

Lead class discussion on findings.

Slide 9: STEM Challenge

10 min

Say: *"Now apply what you learned to design a better compost system."*

Watch For

Distribute design materials, circulate to assist.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Dead Organic Matter, Soil Moisture
- **INTERNAL:** Decomposer Activity, Nutrient Release

Relationships

- **Dead Organic Matter to Decomposer Activity:** POSITIVE (+) — More dead stuff gives decomposers more food to process.
- **Soil Moisture to Decomposer Activity:** POSITIVE (+) — Decomposers need water to survive and work.
- **Decomposer Activity to Nutrient Release:** POSITIVE (+) — More decomposition means more nutrients released.

Key Discoveries

- Decomposers need both food (organic matter) AND moisture to work
- When conditions are right, nutrient release keeps pace with dead matter input
- Drought causes dead matter to pile up because decomposers can't work
- The system recovers when moisture returns, but not instantly

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Decomposers only work on dead animals"	Decomposers break down ALL organic material including leaves, wood, and food waste.
"Dirt is just dirt"	Healthy soil is a living ecosystem with billions of organisms per teaspoon.
"Decomposition destroys matter"	Matter is transformed, not destroyed. Nutrients are released back into the ecosystem.